

Fairness in IRE Notes

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1 Introduction

When a user initiates a search request, the information retrieval (IR) system must efficiently explore the search space and return a set of relevant results. Since relevance is often considered the most important metric of the system, the result set may not guarantee enough diversity coverage. Items in the largest or the most prominent subtopic/aspect group may take up most of the search results, thus leaving little space for the minority aspects to surface. Consequently, it may take a user much more effort to reach an item of a minority aspect of a query topic, leading him/her to a partial—and often very skewed—discovery of information. Furthermore, because users frequently only click on the top results, a ranking or recommendation algorithm that takes user feedback into account, will continue putting those same items at the top, creating a positive feedback loop and a vicious cycle of unfairness (Gao et al., 2022). Now, minority in movie recommendation systems could be a particular genre of movies that are not watched by majority, news recommendation systems could be a particular political leaning/country/ideology, product recommendation system could be a particular company or stuff. Subsequently, if we are summarizing opinions related to a news event that consider different demographic groups, such as gender, then a fair IRE system should ensure equal representation of both the genders in the summary. A demographic group could be based on any private information of an user, such as, religion, region, language, political leaning, caste, gender, ethnicity, country, etc.

Therefore, the basic notion of fairness could be summed up as counterfactual fairness. This captures the intuition that a decision is fair towards an individual if it is the same in (a) the actual world and (b) a counterfactual world where the individual belonged to a different demographic group (Kusner

et al., 2017) (read till section 2.1, for more details). So, for example, in IRE based ranking, if the demographic factor of the particular option shown is changed leads to change in its ranking, then the system developed is biased wrt to *counterfactual fairness*. This applies to any form of recommendation/ summarization and other applications. Based on the topics taught in class, next we will discuss few metrics that could aid us in understanding if the ranking is biased.

We will start with very basic distance based scoring, *Kendall Tau Distance*¹. It finds the distance between two ranked lists, for example, one provided by the proposed IRE system and the other ground truth. The distance, D_{kT} is calculated as

$$D_{kT} = \frac{\text{mismatch} * 2}{N * (N - 1)} \quad (1)$$

where, N represents the number of items in the ranked list and mismatch is calculated as the number of pair of items whose rank differs from the proposed IRE system from ground truth. Suppose, XYZ is a biased news recommender system which provides higher ranks to only P1 leaning based news and GroundTruth represents the actual ranking based on user popularity as shown in Table 1

Table 1: Kendall Tau Example

NewsArticle	XYZ	GroundTruth
N ₁	1	2
N ₂	3	1
N ₃	2	3

Therefore, the distance is calculated as follows, as shown in Table 2

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$$D_{kT} = \frac{2 * 2}{3 * (3 - 1)} \quad (2)$$

¹https://en.wikipedia.org/wiki/Kendall_tau_distance

Table 2: Kendall Tau Example : Calculation

Pairs NewsArticle	XYZ	GroundTruth	MisMatch
(N ₁ , N ₂)	1<3	2>1	Yes
(N ₁ , N ₃)	1<2	2<3	No
(N ₂ , N ₃)	3>2	1<3	Yes

where, number of mismatches shows the number of pairs of items that are ranked differently wrt ground truth. This metric only focuss on the order and does not consider how distant the difference is during mismatches irrespective of maintaining same order. Please refer to the corresponding wikipedia page for more details.

Rank Parity : Since Kendall Tau distance only captures the difference in order wrt items to understand fairness, *Rank Parity* ensures understanding of exposure of items belonging to a particular group, calculated as follows. Each document, d_i at rank (i) gets exposure based on position bias, calculated as follows

$$Exposure(d_i) = \frac{1}{\log_2(1+i)} \quad (3)$$

For a group (G), the exposure depends on the cumulative scores of all the documents that belong to this particular group :

$$Exposure(G) = \sum_{i \in G} Exposure(d_i) \quad (4)$$

Therefore, if we consider an application where we are summarizing user opinions related to an event, E which is gender based, and considering only two corresponding groups that represent the binary genders, *Male* and *Female*. Then, if the tweets shown in ranked order belongs to both the groups wrt to $Exposure(G)$ where G_1 and G_2 represents both the genders respectively, then we can say, our IRE system (maybe, LexRank/MMR/any summarization tool) is fair in this application. This can also be measured by *Rank Parity Difference* of the event E as $RParityD(E)$, as shown next, i.e., difference between average exposure of each group. $Exposure(G_1)$ is represented as $Expre(G_1)$ and similarly for G_2

$$RParityD(E) = \frac{Expre(G_1)}{|Expre(G_1)|} - \frac{Expre(G_2)}{|Expre(G_2)|} \quad (5)$$

IRE based metrics : Few well known metrics that could be modified for fairness in IRE include, **KL divergence and NDCG**. **KL Divergence**

: please refer to any available resource, including ². Similarly, for normalized discounted cumulative gain, refer to ^{3, 4}

Expected Reciprocal Rank, ERR (Chapelle et al., 2009) measures that if an user scans every document from top to bottom and will only stop at the first relevant result else continues. So, its measured by considering $P(stop_r)$ i.e., probability of the user to stop at rank r where n is the number of documents in the ranking. This only means that that the user has not found any relevant document, till $(r-1)$ rank, which is calculated as, $\prod_{i=1}^{r-1} (1 - R(i))$ where $R(i)$ is the relevance score of the i^{th} option/document shown.

$$R(i) = \frac{2^{rel(i)} - 1}{2^{maxrel}} \quad (6)$$

and $rel(i)$ is the relevance score of the document at the i^{th} position and $maxrel$ represents the maximum relevance score among all documents, based on this ERR is calculated as

$$ERR = \sum_{i=1}^n \frac{P(stop_r)}{r} \quad (7)$$

References

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- Matt J Kusner, Joshua Loftus, Chris Russell, and Ricardo Silva. 2017. Counterfactual fairness. *Advances in neural information processing systems*, 30.

²<https://towardsdatascience.com/understanding-kl-divergence-f3ddc8dff254/>

³https://en.wikipedia.org/wiki/Discounted_cumulative_gain

⁴<https://medium.com/@mukulranjan/understanding-ndcg-885656321b3b>